

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
IVth Semester of MSc Biotechnology Examination-April-2019

BT862: Developmental Genetics

Date: 29/4/2019 Day: Monday Time: 1.30 p.m. To 1.50 p.m. Maximum Marks: 10

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I Choose/write the correct answer for the following questions.

10

- 1. Which type of cells are involved in cellular regeneration post apoptosis?**
(i) Adult stem cells (ii) Multipotent stem cells
(iii) Pluripotent stem cells (iv) All of above
- 2. From the following which is directly linked to differentiation of embryonic stem cells?**
(i) Endoderm (ii) Mesoderm
(iii) Ectoderm (iv) All of above
- 3. The outermost layer of fertilized embryo can be termed as**
(i) Trophoblast (ii) Ectoderm (iii) Cell wall (iv) All of above
- 4. In aging cell the destroyed cellular population can be replenish , with respect to this which statement is correct**
(i) Cannot regenerate
(ii) It regenerates through adult stem cells
(iii) It regenerates up to certain age only
(iv) Regeneration process is very slow
- 5. From the following which cellular stage best describes the pre-implantation blastocyst stage?**
(i) Attached blastocyst (ii) 8 cell embryo (iii) Blastocyst free from uterine lining (iv) Blastocyst with inner cell mass

- 6. From the following which factor/s contribute to the transcription complexity**
 (i) Chemical modification (ii) Splicing (iii) Capping (iv) All of above
- 7. In experiment performed towards the aging process in mice revealed**
 1. There is change in relative methyl cytosine content with aging
 2. There would be no tissue specific change in methylation content
 (i) 1 & 2 (ii) Only 1 (iii) Only 2 (iv) None of above
- 8. The nuclei of essentially all differentiated adult cells of an individual are genetically (though not necessarily metabolically) identical to one another and to the nucleus of the zygote from which they descended this can be termed as**
 (i) Multipotancy (ii) Nuclear equivalence (iii) pluripotent (iv) None of above
- 9. Which of the following is key factor for inducing the genetic mosaicism ?**
 (i) Differential expression of genes (ii) Somatic mutation on the genes
 (iii) RNA interference (iv) All of above
- 10. A non-sequence dependent inheritance can be explained through**
 (i) Dominance and recessive interaction (ii) Imprinted alleles
 (iii) Epigenetic interaction of allele (iv) Both (ii)&(iii)

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Date: 29/4/2019 Day: Monday Time: 1.50 p.m. To 3.30 p.m. Maximum Marks: 25

Instructions:

- 1. Section I and II must be attempted in TWO ANSWER SHEET.**
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION -I

Q-II Answer the following questions as directed [Any Two] 10

- 1** What is DNA hypo methylation ?Discuss the role of DNA methylation in aging and inducing age-related disorders
- 2** What is RISC? What is DICER? Present a mechanism of RISC and DICER mediated method of gene silencing.
- 3** Discuss various types of stem cells. Explain the method for derivation of Embryonic stem cells.

SECTION - II

Q-IIIA Answer any two of the following 10

- 1** What is epigenetic reprogramming give the mechanism of reprogramming with suitable example
- 2** How many types of methyl transferase are known to you? What are the differences in their function? Describe its association with SAM? Name any two methods that can be used to detect methylation
- 3** What is epigenetics? How does it gain control over the gene expression? Name two key pathways that facilitate its functioning. Describe its role in brief

Q-IIIB Write a note on any one of the following: 5

- 1** Discuss the mechanism of epigenetic imprinting with suitable example
- 2** How does the methylation changes occur during the development of embryo